

Arghya Maity

Curriculum Vitae

Webpage: Google Scholar

Personal Details:

Name: Arghya Maity
Nationality: Indian
DOB: 04 May, 1989
Position: Postdoctoral Researcher
Address: Harish-Chandra Research Institute (HRI),
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Gender: Male
Marital Status: Married
Language: Bengali, Hindi, English



Educational Qualifications:

Degree	University	Year & Results	Subjects
Postdoc.	HRI, QIC Group Prayagraj, India	Since, 8th May,2021	Quantum Information Theory Quantum Thermodynamics
Postdoc.	IIT-Bombay Mumbai, India	6th Nov,2020 to 7th May,2021	Theoretical Physics Soft Condensed Matter
Ph.D.	BITS-Pilani, Department of Physics, Pilani, India	2014 to 2020	Theoretical Physics Soft Condensed Matter
M.Sc.	Sambalpur University, Sambalpur Odisha, India	2013 7.56(OGPA) $\equiv$ 71% 1 st class	Physics Special paper: High Energy Physics
B.Sc. (Honours)	Vidyasagar University, Midnapur West Bengal, India	2010 52 % (2 nd class)	Physics (Hons.) Mathematics (General), Chemistry (General) Bengali, English (compulsory)
H.S. (12 th)	W.B.C.H.S.E. West Bengal, India	2006 60 % (1 st class)	Pure Science, Bengali, English (compulsory)
10 th	W.B.B.S.E. West Bengal, India	2004 71 % (1 st class)	Bengali-I,Bengali-II, English, Math, PSC, LSC, History, Geo

List of Publications/Preprints: Webpage: Google Scholar

Not listing the papers in preparation. All papers are not available in arXiv.

Research Article:

13. **Arghya Maity**, P. Chaki, Ahana Ghoshal, Ujjwal Sen; "Quantum Heat Transformers", arXiv:2404.00584 (2024), [Link](#)
12. **Arghya Maity**, Ahana Ghoshal; "Using non-Markovian dynamics in effective negative temperature based transient quantum Otto engines", Physical Review A **109**, 022207 (2024), ([Editors' Suggestion](#)), [Link](#)
11. **Arghya Maity**, Aditi Sen De; "Enhanced efficiency in quantum Otto engine via additional magnetic field and effective negative temperature", arXiv:2304.10420 (2023), [Link](#)
10. T. K. Konar, **Arghya Maity**, Aditi Sen De; "Classical correlations for Generic States are Fragile under Decoherence", arXiv:2209.03334 (2022), [Link](#)
9. A. Patra, **Arghya Maity**, Aditi Sen De; "Resource Theory of Non-absolute Separability", Physical Review A **108**, 042402 (2023), [Link](#)
8. R. Gupta, **Arghya Maity**, S. Mal, Aditi Sen De ; "Statistics of Entanglement Transformation with Hierarchies among Catalysts", Physical Review A **106**, 052402 (2022), [Link](#)
7. **Arghya Maity**, N. Mathur, Petra Imhof, Navin Singh ; "Structural Analysis of DNA molecule in a confined shell", Physica A: Statistical Mechanics and its Applications **609**(128382), 0378-4371 (2023) [Link](#)
6. **Arghya Maity**, Navin Singh, "Melting of DNA in confined geometries", European Biophysics Journal, **49** (7) 561-569 (2020) , [Link](#)
5. **Arghya Maity**, A. Singh, Navin Singh, "Stability of DNA passing through different geometrical pores", EPL(Europhysics Letters), **127** (2) 28001 (2019) , [Link](#)
4. **Arghya Maity**, A Singh, Navin Singh, "Differential stability of DNA based on salt concentration", European Biophysics Journal; **46** (1) 33-40 (2017) , [Link](#)
3. **Arghya Maity**, A Singh, Navin Singh, "DNA denaturation in ionic solution", American Institute of Physics (AIP) Conference Proceedings; **1728**, 020415 (2016), [Link](#)

Review Article:

2. A. Singh, **Arghya Maity**, N. Singh ; "Review: Structure and dynamics of dsDNA in cell mimicking environments", Entropy, **24**(11),1587 (2022) [Link](#)

Book Chapters:

1. N. Mathur, **Arghya Maity**, N. Singh, Booktitle : "Advances in Computational Modeling and Simulation", Publisher : "**Springer Nature Singapore**", Chapter Title : "DNA Molecule Confined in a Cylindrical Shell: Effect of Partial Confinement", isbn="978-981-16-7857-8" (2022), [Link](#)

Skills:

- **Programming Languages: Efficient in C, C++ and Python**
- Software Packages : Efficient in **Mathematica, QuTiP** (Quantum Toolbox in Python)
- **Molecular Dynamics (MD) Simulation** - GROMACS
- Software Packages : Gnuplot, xmgrace, LATEX, VMD(Visual Molecular Dynamics), Curves+
- Operating Systems: Efficient in Linux.
- Server Maintenance

Awards, Fellowships:

1. Sandwich Research Scholarship: Freie Universitat Berlin, Germany, (2018) (During Ph.D).
2. Mexican Council of Science (CONACyT) PhD Scholarship,(2015) (Full time Ph.D. Scholarship).

Research Experience:

1. Working as Research Associate in "**QuEST**" - **Quantum-Enabled Science & Technology**, project, (8th May, 2021 to 4th Oct, 2023), sponsored by "The Department of Science & Technology (DST)", INDIA.
2. Guest Researcher in Department of Physics at Free University of Berlin, Berlin, Germany,(4th Year of Ph.D.).
3. Project Fellow (JRF(Junior Research Fellow) and SRF (Senior Research Fellow)) of "The Department of Science & Technology (DST)", INDIA, at BITS-Pilani.
4. Short term Project : "Thin Film Processing Techniques and application of α -spectroscopy to measure the thickness of the Film." at UGC-DAE CONSORTIUM FOR SCIENTIFIC RESEARCH,Kolkata, INDIA (M.Sc Project- April 2012).

Teaching Assistantships:

- Teaching Assistant(TA): **NPTEL(National Programme on Technology Enhanced Learning)** (Initiated by IITs and IISC), (2021).
- Teaching Assistant(TA) with Prof.Ranjith Padinhateeri at **IIT-Bombay**,(2021).
- Teaching Assistant (TA) in Physics Lab Course for 1st year B.Tech Student at **BITS-Pilani**, Pilani Campus, Department of Physics, (2014-2019, 5 years).
- Guest Lecturer for a Government Engineering College , "**VSSUT (Veer Surendra Sai University of Technology)**", Sambalpur, Burla, Odisha, INDIA,(July 2013 - Nov 2013).

Presentation in Conferences:

1. Young Quantum - 2023 (YouQu-2023), organized by the Quantum Information and Computation group(QIP) of Harish-Chandra Research Institute (HRI), Prayagraj (Allahabad), 15-18 Feb, 2023. Presented a **talk** on “*Resource Theory of Non-absolute Separability*” (in person).
2. Puri Polymer Conference 2018, organized by the Institute of Physics, Bhubaneswar, 12-14 Dec, 2018. Presented a **talk** on “*Stability of DNA within the matrix of different geometric confinement*” (in person).
3. American Institute of Physics (AIP) Conference Proceedings, International Conference on Condensed Matter & Applied Physics (ICC), October 2015. Presented a **poster** on “*DNA Denaturation in Ionic Solution*” (in person).

Professional Memberships/Academic Responsibilities:

- **Member of the organizing committees:** QIPA-2023 (International Conference), held at the Harish-Chandra Research Institute, India.
- **Member of the organizing committees:** Young Quantum 2023 (International Meeting), held at the Harish-Chandra Research Institute, India.
- **Member of the organizing committees:** DNA Physics-2017 (International Conference), held at the BITS-Pilani, Pilani Campus, India.

Research Interests:

My areas of interest have expanded over time. I started my research journey with Statistical Thermodynamics, Soft Condensed Matter, Molecular Dynamics (MD), and then I moved toward Quantum Thermodynamics, Open Quantum systems, and Quantum Information Theory. Listed below are a few sub-areas in which I have spent my time till now:

- Quantum Thermodynamics.
- Open Quantum System : Markovian & non-Markovian Dynamics, Collision models
- Quantum Thermal Devices : Quantum Engine, Quantum Refrigerator, Quantum Battery, Quantum Transistor
- Resource Theory : Catalytic Resource theory.
- Different Quantum and Classical Correlations with Decoherence.
- Quantum Metrology : Quantum Thermometry.
- Quantum Jarzynski Equality in Open Quantum Systems.
- Quantum Biology : Quantum Coherence and Entanglement in the Avian Compass: Avian Magnetoreception through ”radical pair” Model, Biological Heat Engines.
- Simulation : Molecular Dynamics (Simulation).
- Soft Condensed Matter : Thermodynamics of bio-molecules.

References:

1. Prof. Navin Singh

Professor

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2. Prof. Aditi Sen De

Professor H, Physics

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3. Prof. Ujjwal Sen

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